

constitution_advance_investigation

Constitution-submodule advance investigation (§11.4.71 / §11.4.37 / §11.4.55)

- **Date:** 2026-06-24
- **Mode:** READ-ONLY investigation. No pull, no gitlink bump, no cascade. Submodule HEAD left at 1d9e3ce.
- **Recorded gitlink (meta-repo):** 1d9e3ce (1d9e3cecf209315273b06beb4cfb7d1013f0f32b)
- **Upstream github/main (fetched):** e1bb125 (e1bb12502d297ccef376698fc2cadd6a92d2b112)
- **Submodule path:** ./constitution (repo root — NOT submodules/constitution)

1. New commits 1d9e3ce..e1bb125 (oldest→newest)

SHA	Subject	One-line effect
bbe84e3	test(anti-bluff): scanner-triage — real observable assertions + honored skip annotations (§11.4)	Test-hardening INSIDE the constitution submodule's own scripts/workable-items/*_test.go (drove anti-bluff-scan survivors 140→13). No mandate change, no thresholds weakened. No consumer action.
c670154	feat(§11.4.167,§11.4.168): feature-work-stream lifecycle + visual-doc-validation mandates + §11.4.157 carrier back-fill	Adds TWO new universal mandates (§11.4.167, §11.4.168) to Constitution.md + mirrors them into the submodule's own CLAUDE/AGENTS/QWEN/GEMINI carriers. This is the only commit that introduces new rules.
b2a7948	feat(§11.4.73): author default-md.css + default-pdf.css project styling stylesheets	Authors styles/default-md.css + styles/default-pdf.css (referenced by every consumer's export pipeline via --css/--stylesheet but previously missing → unstyled exports + weasyprint FileNotFoundError). Lives in the constitution submodule;

SHA	Subject	One-line effect
e1bb125	docs(§11.4.65/§11.4.73): regenerate constitution carrier siblings (html/pdf/docx) with default CSS	consumed by reference (§11.4.80-style). Pure regen of the submodule's OWN *.html/*.pdf/*.docx siblings with the new CSS. Doc-export hygiene inside the submodule. No consumer action.

2. New mandates + required consumer (HelixCode) actions

§11.4.167 — Big-work-item feature work-stream lifecycle (User mandate 2026-06-23)

Every BIG work item (new feature OR drastic/large fix) MUST be its own isolated feature work-stream: own full project copy in a sibling dir (<project>_<slug>), own feature/<slug> branch, own per-feature builds + feature tags, kept SEPARATE from trunk until operator manually approves after testing, with trunk merged INTO every stream regularly. Sub-clauses (A)–(H): (A) one big item = one sibling copy, small changes stay on trunk; (B) disk-feasible CoW/reflink clone MANDATORY (cp -a --reflink=auto on btrfs/XFS/ZFS/APFS, else git worktree + externalized build output), naive deep copies / git clone per stream FORBIDDEN, ~0-disk verified by captured evidence before relied on; (C) own branch + greppable tags (<base>-feat-<slug>[-<iter>] per §11.4.151), NEVER merged to trunk until §11.4.40 full retest GREEN on clean target → §11.4.41 merge-first → §11.4.113 ff-only; (D) trunk merged INTO every live stream per-trunk-tag/daily (git merge, never rebase a shared/tagged branch); (E) feature branch + mirrored feature tag cascades to EVERY touched submodule the moment it is modified, untouched submodules stay trunk-pinned; (F) single-builder FIFO build queue + finite-device test queue (§11.4.119 per-device exclusive ownership), idle out/ GC'd; (G) every change crosses the full review/impact/anti-bluff gauntlet — a stream is NOT a quality carve-out; (H) resumable, data-safe lifecycle via a stream registry + atomic .partial→rename create + §9.2 retire-backup. Universal (§11.4.17); consumer fills CoW mechanism, naming, queue counts, big-item threshold per §11.4.35. Recommended gates: CM-FEATURE-WORKSTREAM-COW-CLONE / -NO-MERGE-UNTIL-APPROVED / -TRUNK-SYNC-CADENCE / -SUBMODULE-CASCADE.

HelixCode required action: (1) cascade the compact 11.4.167 mirror into root CLAUDE.md/AGENTS.md/GEMINI.md/QWEN.md (§11.4.157/§11.4.26 — they currently top out at 11.4.165); (2) eventually wire the four recommended gates + paired §1.1 mutations (separate gate-code work item). Note: macOS host → APFS supports cp -c clonefile / cp -a --reflink is not the macOS flag — the consumer §11.4.35 instantiation must pick the APFS CoW mechanism (cp -c clonefile or APFS snapshot).

§11.4.168 — Exported-document independent content + textual + full-visual validation (User mandate 2026-06-23)

Every generated/exported document (HTML/PDF/DOCX/any §11.4.65 export-scope format) MUST pass INDEPENDENT validation by a reviewer structurally separate from the generator (NEVER author self-check), on BOTH source AND every exported artifact, ALWAYS, across THREE layers: (1) CONTENT — export faithfully carries source intent+data, nothing dropped/truncated/garbled; (2) TEXTUAL — human-readable, NO raw markup / diagram-source (Mermaid gantt/graph/flowchart/sequenceDiagram/...)/unrendered code-fence leaking as body text; (3) FULL VISUAL — diagrams render as IMAGES not source, layout intact, verified by RENDERING the export (pdftotext catches raw-source-as-text + pdfimages confirms rendered images + pdftoppm→OCR confirms human-readable content) with captured evidence (§11.4.5/.69/.107). Analyzer self-validated golden-good/golden-bad (§11.4.107(10)); iterate to clean GO (§11.4.134). Forensic anchor: 2026-06-23 raw-gantt-in-PDF leak the green suite missed (§11.4.138 operator-escape) — `class="mermaid"` HTML grep is a SOURCE check, NOT an exported-artifact check. Honest boundary: confirms exported artifact is faithful/readable/rendered — does NOT prove source content correct nor replace the §11.4.65 mtime-parity freshness gate. Universal (§11.4.17). Recommended gate: CM-EXPORTED-DOC-VISUALLY-VALIDATED.

HelixCode required action: (1) cascade the compact 11.4.168 mirror into the four root carriers (§11.4.157/§11.4.26); (2) eventually wire CM-EXPORTED-DOC-VISUALLY-VALIDATED + paired §1.1 mutation, and apply the three-layer validation to HelixCode's own doc-export pipeline (Status/Status_Summary/feature-status HTML/PDF/DOCX). HelixCode does have a doc-export/QA surface, so this binds — but as wiring, not as a blocker on the current code/test cycle.

§11.4.73 CSS authoring (styles/default-md.css + styles/default-pdf.css)

Not a new mandate — fills a long-standing gap. Consumed by reference at constitution/styles/*.css (NOT copied, §11.4.80-style). Affects HelixCode only if/when HelixCode runs the constitution-driven doc-export pipeline pointing `--css/--stylesheet` at these files.

3. Classification of the advance

- **(a) Anchors needing only consumer-fleet propagation-literal added (cascade):** §11.4.167 and §11.4.168 — add the compact mirror (literal 11.4.167 / 11.4.168) to root CLAUDE.md/AGENTS.md/GEMINI.md/QWEN.md. Currently ALL FOUR carriers top out at 11.4.165 → uniform 2-anchor gap (confirmed: 167=0 168=0 in every carrier).
- **(b) New enforceable gates the consumer must wire:** CM-COVENANT-114-167-PROPAGATION, CM-COVENANT-114-168-PROPAGATION (propagation literals), plus recommended functional gates CM-FEATURE-WORKSTREAM-COW-CLONE / -NO-MERGE-UNTIL-APPROVED / -TRUNK-SYNC-CADENCE / -SUBMODULE-CASCADE (§11.4.167) and CM-EXPORTED-DOC-VISUALLY-VALIDATED (§11.4.168). Each is its own gate-code work item with paired §1.1 mutation.
- **(c) Informational / no consumer action:** bbe84e3 (submodule-internal test triage), e1bb125 (submodule-internal sibling regen). b2a7948 is consumed-by-

reference infra (action only if HelixCode invokes the constitution export pipeline).

- **(d) Anything that changes how the CURRENT build/test cycle should run: NONE.** Neither §11.4.167 nor §11.4.168 alters HelixCode's build/boot/runtime expectations or the current code/test cycle's pass criteria. §11.4.167 governs how FUTURE big work items are organised (branch/clone/queue discipline) — it does not retroactively invalidate the in-flight cycle. §11.4.168 governs doc-export validation — orthogonal to the code/test cycle (it would add a doc-export gate, not change test outcomes).

4. Recommendation

None of this must be done BEFORE the current test cycle. The advance contains:
- 2 new universal governance mandates requiring a **cascade** into HelixCode's four root carriers (§11.4.157/§11.4.26) + future gate-wiring, - 2 submodule-internal hygiene commits (no consumer action), - 1 consumed-by-reference CSS-infra commit.

Per §11.4.121 (no mid-cycle change to avoid racing the in-flight cycle) and §11.4.92 multi-pass, the gitlink-bump + cascade should be a TRACKED FOLLOW-UP cycle the conductor sequences AFTER the current test cycle completes — NOT interleaved now. Flag check: no new rule changes test/boot expectations, so deferring is safe.

The follow-up cycle (conductor-sequenced) would: fetch+investigate (done here) → §11.4.113 ff-only update of the submodule to e1bb125 (no force) → bump the meta-repo .gitmodules/gitlink pointer in the SAME commit as the cascade (§11.4.49 step 7) → add the 11.4.167/11.4.168 compact mirrors to CLAUDE.md/AGENTS.md/GEMINI.md/QWEN.md → run scripts/verify-all-constitution-rules.sh (§11.4.55) → wire the recommended gates as separate gate-code items.

Submodule HEAD verified untouched at 1d9e3ce (fetch-only; no checkout).